

DATABASE MANAGEMENT SYSTEM

Unit-IV: MS Access

Comprehensive Study Notes with Practical Concepts

UNIT-IV: MS ACCESS

Topic 1: Introduction to MS Access

1.1 What is MS Access?

Microsoft Access is a **Relational Database Management System (RDBMS)** that combines the relational Microsoft Jet Database Engine with a graphical user interface and software-development tools. It is part of the Microsoft Office suite and is designed for small to medium-sized businesses and individual users.

MS ACCESS - KEY FEATURES	
MICROSOFT ACCESS	
Tables	Store data in rows and columns
Queries	Retrieve and manipulate data
Forms	User interface for data entry
Reports	Print formatted data
Macros	Automate tasks
Modules	VBA programming

1.2 Benefits of MS Access

Benefit	Description
User-Friendly	Easy-to-use graphical interface
Rapid Development	Quick to create databases and applications
Integration	Seamless integration with other Office applications
Template Support	Built-in templates for common database needs
Scalability	Suitable for small to medium-sized databases
Report Generation	Professional reporting tools

Security	User-level security and password protection
Cost-Effective	Part of Microsoft Office suite

Topic 2: Tables

2.1 What are Tables?

A **table** is the fundamental structure in MS Access that stores data in a relational format. Tables consist of **fields** (columns) and **records** (rows).

TABLE STRUCTURE IN MS ACCESS			
Table: STUDENT			
StudentID (PK)	Name	Age	Department
S001	Rahul Sharma	20	Computer Science
S002	Priya Singh	21	Mathematics
S003	Amit Verma	19	Computer Science
↑ Field	↑ Field	↑ Field	↑ Field

2.2 Creating Tables

2.2.1 Design View (Creation/Design Structure)

Steps:

1. Open MS Access
2. Click **Create** tab → **Table Design**
3. Enter field names
4. Select data types
5. Set properties
6. Define primary key
7. Save the table

TABLE DESIGN VIEW

Table Design View		
Field Name	Data Type	Description
StudentID	AutoNumber	Primary Key
Name	Text	Student Name
Age	Number	Age of Student
Department	Text	Department Name

2.2.2 Data Types in MS Access

Data Type	Description	Example
Short Text	Up to 255 characters	"John Smith", "New Delhi"
Long Text	Up to 64,000 characters	Long descriptions, notes
Number	Numeric values	25, 100.5
Large Number	Large numeric values	1,000,000,000
Date/Time	Date and time	15-Jul-2026, 10:30 AM
Currency	Monetary values	₹1,000.00, \$50.50
AutoNumber	Automatically generated number	1, 2, 3, 4...
Yes/No	Boolean values	Yes/No, True/False
OLE Object	Images, documents, files	Pictures, Word documents
Hyperlink	Web addresses	https://example.com
Attachment	Multiple files per record	Multiple attachments
Calculated	Calculated field	=[Price]*[Quantity]
Lookup Wizard	Look up values from another table	Department names

2.2.3 Data Entry in Tables

Methods:

1. **Datasheet View:** Double-click the table → Enter data directly into cells → Press Tab to move to the next field
2. **Form View:** Create a form → Enter data in the form interface
3. **Import Data:** Import from Excel, CSV, XML, etc.

2.3 Field Properties

Property	Description
Field Size	Maximum characters allowed
Format	Display format for the field
Input Mask	Template for data entry
Caption	Label for the field
Default Value	Value automatically entered
Validation Rule	Rule to validate data
Validation Text	Error message for validation
Required	Must have a value
Allow Zero Length	Allow empty strings
Indexed	Yes (No Duplicates) / Yes (Duplicates OK)

2.4 Primary Keys

Definition: A **Primary Key** is a field or combination of fields that uniquely identifies each record in a table.

Characteristics:

- Must be unique for each record
- Cannot be null
- Only one primary key per table

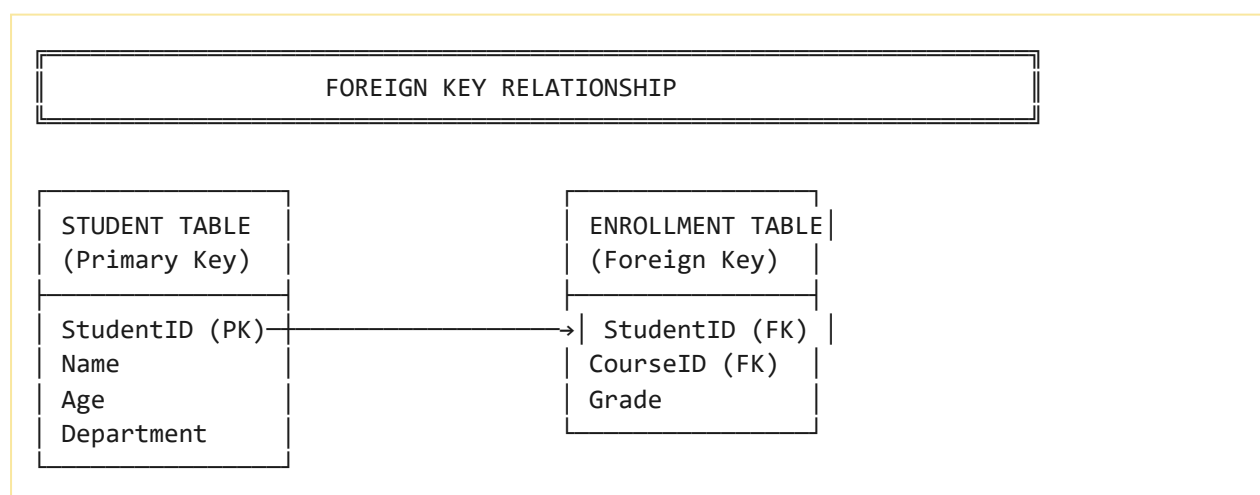
Type	Description	Example
Single-Field Key	One field as primary key	StudentID
Composite Key	Multiple fields together	StudentID + CourseID
AutoNumber Key	Automatically generated	1, 2, 3...

Setting Primary Key:

1. Open table in Design View
2. Select the field(s)
3. Click **Primary Key** button in ribbon

2.5 Foreign Keys

Definition: A **Foreign Key** is a field in one table that references the Primary Key of another table. It establishes relationships between tables.

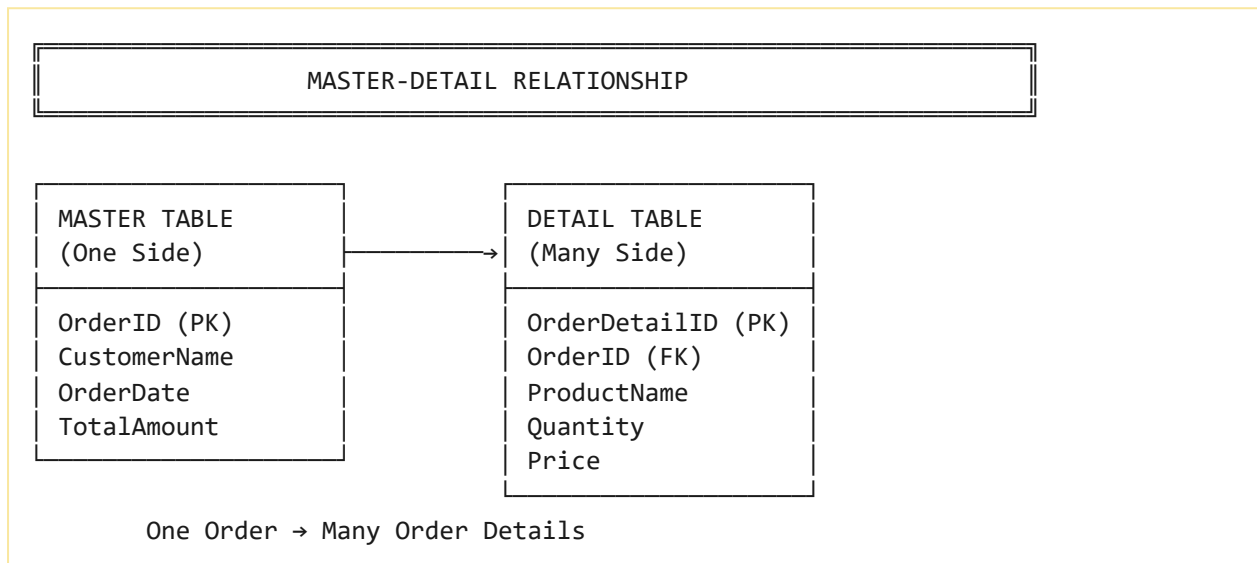


Setting Foreign Key:

1. Create relationship between two tables
2. Link the foreign key field to the primary key field in the parent table

2.6 Master-Detail Table

Definition: A **Master-Detail relationship** exists between two tables where one table is the "master" (one side) and the other is the "detail" (many side).

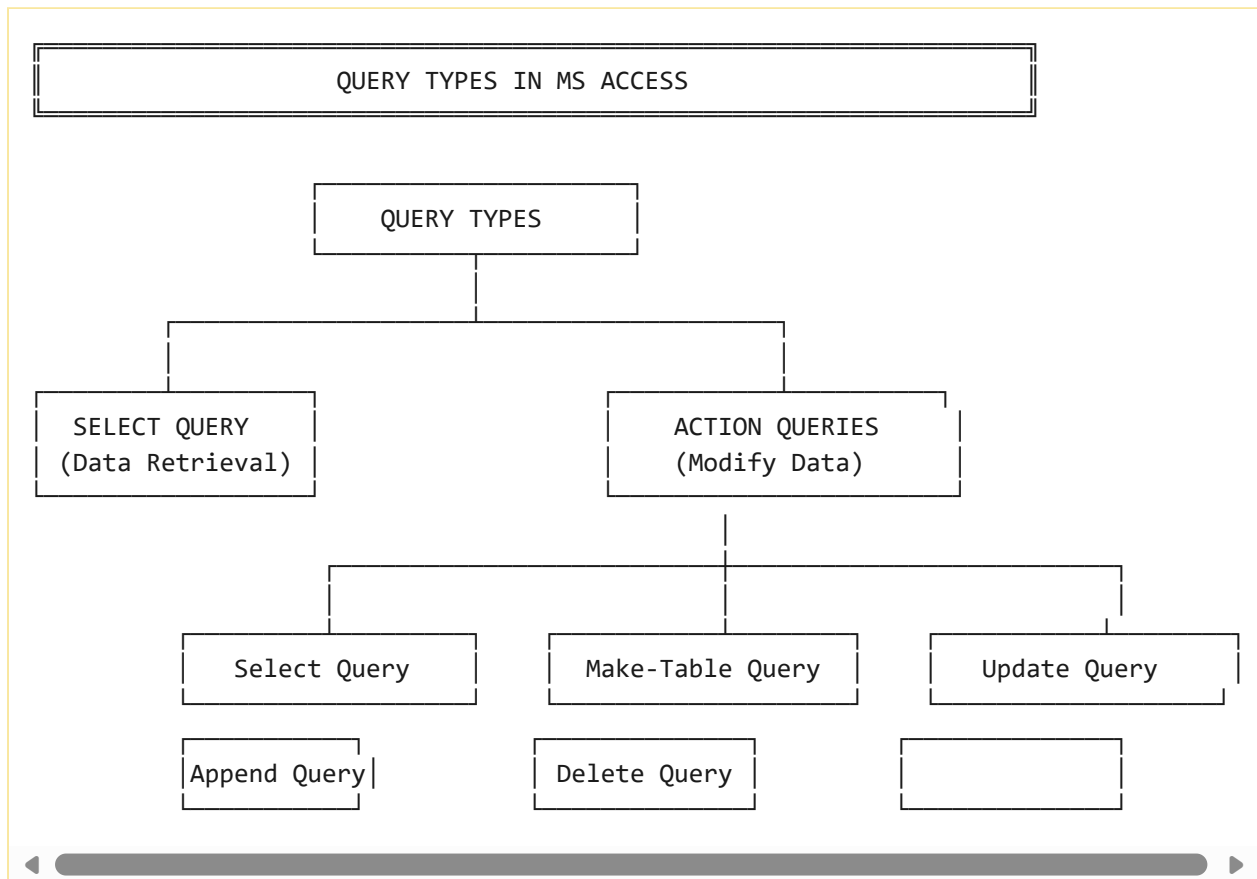


Example: One customer can have many orders (One-to-Many relationship).

Topic 3: Queries

3.1 What is a Query?

A **query** is a request for data from a database. It allows you to retrieve, update, delete, or append data based on specific criteria.



3.2 Select Query

Purpose: Retrieves data from one or more tables based on specified criteria.

Steps to Create Select Query:

1. Click **Create** tab → **Query Design**
2. Add tables
3. Select fields
4. Set criteria
5. Run the query

Example:

```
SELECT StudentID, Name, Age
FROM STUDENT
WHERE Age > 20;
```

QUERY DESIGN VIEW

Query Design View

Field:	StudentID	Name	Age
Table:	Student	Student	Student
Sort:	Ascending		
Show:	☒	☒	☒
Criteria:			>20

3.3 Make-Table Query

Purpose: Creates a new table from the results of a query.

Steps:

1. Create a Select Query
2. Click **Query Type** → **Make Table**
3. Enter the new table name
4. Run the query

Example: Create a new table "SeniorStudents" with students above age 20.

```
SELECT StudentID, Name, Age
INTO SeniorStudents
FROM STUDENT
WHERE Age > 20;
```

3.4 Update Query

Purpose: Updates existing data in a table.

Steps:

1. Create a query

2. Click **Query Type** → **Update**
3. Specify update criteria
4. Run the query

Example: Increase age of all students by 1 year.

```
UPDATE STUDENT  
SET Age = Age + 1  
WHERE Age > 0;
```

3.5 Append Query

Purpose: Adds records from one table to another table.

Steps:

1. Create a query
2. Click **Query Type** → **Append**
3. Select destination table
4. Match fields
5. Run the query

Example: Append new students to the STUDENT table.

```
INSERT INTO STUDENT (StudentID, Name, Age, Department)  
SELECT StudentID, Name, Age, Department  
FROM NewStudents;
```

3.6 Delete Query

Purpose: Deletes records from a table based on specified criteria.

Steps:

1. Create a query
2. Click **Query Type** → **Delete**
3. Specify deletion criteria
4. Run the query

Warning: Delete queries cannot be undone!

Example: Delete students with Age > 25.

```
DELETE FROM STUDENT  
WHERE Age > 25;
```

3.7 Query Comparison Summary

Query Type	Purpose	Data Modified	Can be Undone
Select	View data	No	-
Make-Table	Create new table	Yes (new table)	No
Update	Modify existing data	Yes	No
Append	Add records	Yes	No
Delete	Remove records	Yes	No

Topic 4: Forms

4.1 What are Forms?

A **Form** is a graphical user interface for entering, editing, and viewing data in a database. It provides an easy-to-use interface for data entry.

FORM STRUCTURE

Student Entry Form

Student ID:	[S001	]			
Name:	[Rahul Sharma	]			
Age:	[20	]			
Department:	[Computer Science	]			
[Add Record] [Save Record] [Delete Record] [First] [Previous] [Next] [Last]					

4.2 Types of Forms

Type	Description
Single Form	Displays one record at a time
Continuous Form	Displays multiple records in a list
Datasheet Form	Displays records in a grid format
Split Form	Combines Form and Datasheet views
Main/Subform	Displays master-detail relationship
Navigation Form	Navigation between different forms

4.3 Modal and Modeless Forms

Feature	Modal Form	Modeless Form
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Focus	Must be closed before other windows can be used	Can switch to other windows while open
Interaction	Restricted to the form only	Can interact with other forms
Usage	Error messages, critical dialogs	Data entry, navigation
Examples	Confirm Delete dialog	Main data entry form

Modal Properties:

- **Modal:** Yes → User cannot interact with other forms
- **Modal:** No → User can interact with other forms

4.4 Creating Forms

Methods:

1. **Form Tool (Quick)** - Select table → Click **Create** → **Form** - Auto-generates a form
2. **Form Design** - Click **Create** → **Form Design** - Add controls manually - Customize layout
3. **Form Wizard** - Click **Create** → **Form Wizard** - Follow step-by-step instructions
4. **Blank Form** - Click **Create** → **Blank Form** - Design from scratch

4.5 Form Controls

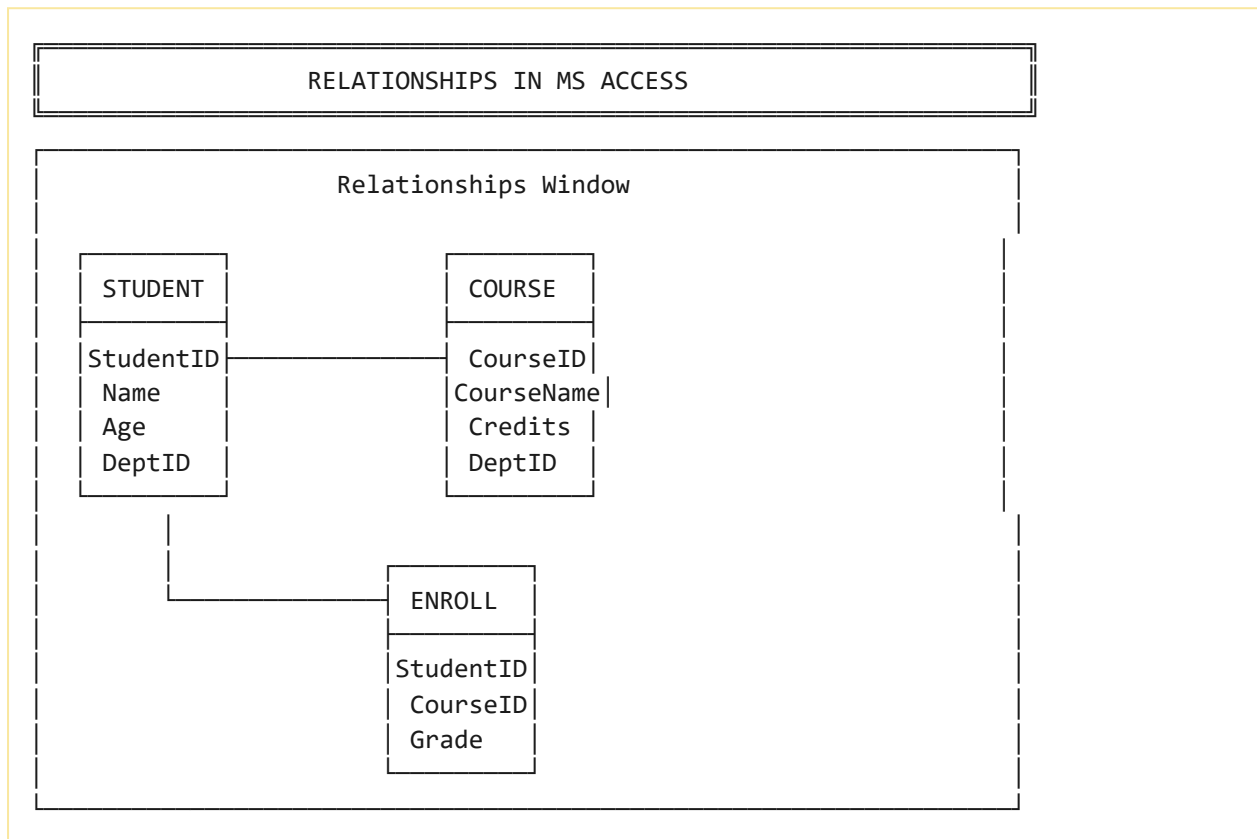
Control	Description
Text Box	Enter or display data
Label	Display text
Button	Execute actions
Combo Box	Drop-down list
List Box	List of items
Check Box	Yes/No values
Option Group	Group of options

Subform	Display related records
Image	Display images

Topic 5: Relationships

5.1 What are Relationships?

Relationships define how tables in a database are connected to each other. They enforce referential integrity and allow data to be combined for queries and reports.



5.2 Types of Relationships

Type	Description	Example
One-to-One	One record in Table A matches one record in Table B	Employee ↔ Person
One-to-Many	One record in Table A matches many records in Table B	Department → Students
Many-to-Many	Many records in Table A match many records in Table B	Students ↔ Courses (via Enrollment)

5.3 Creating Relationships

Steps:

1. Click **Database Tools** → **Relationships**
2. Add tables
3. Drag primary key to foreign key
4. Enforce referential integrity
5. Set cascade options (optional)

Referential Integrity Options:

Option	Description
Enforce Referential Integrity	Prevents orphan records
Cascade Update Related Fields	Automatically updates foreign keys
Cascade Delete Related Records	Automatically deletes related records

Topic 6: Reports

6.1 What are Reports?

A **Report** is a formatted presentation of data used for printing or viewing. It can include grouping, summaries, and calculations.

REPORT EXAMPLE			
STUDENT REPORT			
Department: Computer Science			
StudentID	Name	Age	Courses
S001	Rahul Sharma	20	DBMS, Python
S003	Amit Verma	19	DBMS, Java
Total Students: 2			

6.2 Creating Reports

6.2.1 Simple Report from a Table

Steps:

1. Select the table
2. Click **Create** → **Report**
3. Auto-generated report
4. Modify in Design View

Features:

- All fields from the table are included
- Simple columnar layout
- Pre-formatted

6.2.2 Report from a Query

Steps:

1. Create a query
2. Select the query
3. Click **Create** → **Report**
4. Auto-generated report based on query

Advantages:

- Can filter, sort, and group data
- Data can be from multiple tables
- More customized reports

6.3 Report Design View

Sections in Report Design:

Section	Description
Report Header	Appears once at the beginning
Page Header	Appears at the top of each page
Group Header	Appears at the beginning of each group
Detail	Appears for each record
Group Footer	Appears at the end of each group
Page Footer	Appears at the bottom of each page
Report Footer	Appears once at the end

6.4 Report Controls

Control	Purpose
Text Box	Display field values and calculations
Label	Display static text

Image	Add logo or images
Line	Draw lines
Subreport	Embed another report
Page Numbers	Display page numbers

Topic 7: HPU Important Questions

Short Answer Questions (2-5 Marks)

1. What is MS Access?
2. What are the main objects in MS Access?
3. What is a table in MS Access?
4. What is a primary key?
5. What is a foreign key?
6. What is a query? Name the types.
7. What is a form? What are the types of forms?
8. What is a report?
9. What is a modal form?
10. What are relationships in MS Access?

Long Answer Questions (10 Marks)

1. Explain tables in MS Access with creation and design.
2. Explain primary keys and foreign keys with examples.
3. Explain the different types of queries in MS Access.
4. Explain the master-detail relationship in MS Access.
5. Explain forms in MS Access with creation and types.
6. Explain reports in MS Access with creation and examples.

Practical/Hands-On Questions (10 Marks)

1. Create a table STUDENT with fields: StudentID, Name, Age, Department, Grade. Set StudentID as primary key.
2. Create a query to retrieve all students with Age > 20.
3. Create a form for the STUDENT table.
4. Create a report from the STUDENT table.

5. Create a master-detail relationship between STUDENT and COURSE tables.

Objective Questions (1 Mark)

1. The default file extension of MS Access is:

- (a) .accdb (b) .mdb (c) .xlsx (d) .docx

Answer: (a)

2. Which object is used for data entry?

- (a) Table (b) Query (c) Form (d) Report

Answer: (c)

3. A primary key is:

- (a) Unique (b) Not null (c) Both (d) None

Answer: (c)

4. Which query creates a new table?

- (a) Select (b) Append (c) Make-Table (d) Update

Answer: (c)

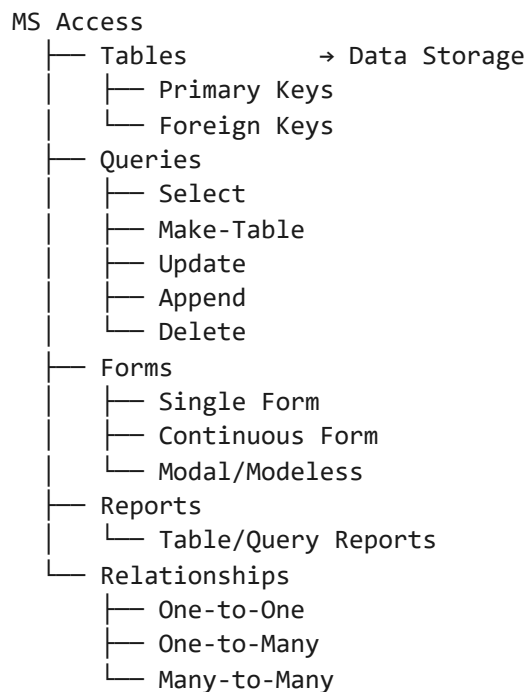
5. A modal form:

- (a) Blocks other windows (b) Allows other windows (c) Both (d) None

Answer: (a)

Quick Revision

MS Access Objects Summary



Key Shortcuts in MS Access

Shortcut	Action
Ctrl+N	New Database
Ctrl+O	Open Database
Ctrl+S	Save
Ctrl+Z	Undo
Ctrl+F	Find
Ctrl+H	Replace

Data Types Summary

Data Type	Use
Short Text	Up to 255 characters
Long Text	Up to 64,000 characters
Number	Numeric values
Date/Time	Dates and times
Currency	Monetary values
AutoNumber	Automatic numbering
Yes/No	Boolean values

End of Unit-IV Notes — DATABASE MANAGEMENT SYSTEM — MS Access